DocShield AI

AI Document Authenticity Screening

SUM 72H Build Challenge · Deliverable 01 — Challenge Summary · Ahmed Hajji · Individual
Live prototype: trust-doc-ai.vercel.app · Source: github.com/Hajji-ahmed/TrustDoc-AI

PROBLEM → SOLUTION → PROTOTYPE → VALIDATION → BUSINESS LOGIC

1. PROBLEM

Official documents are falsified in ways that are hard to catch by eye: edited text, replaced photos, forged stamps, copied signatures, altered QR codes, and increasingly, documents generated from scratch by AI.

Manual verification is slow, repetitive and inconsistent. A reviewer checking dozens of files a day cannot reliably compare font rendering, recompute totals and validate identifiers on every page — and the most damaging forgeries are precisely the subtle ones.

2. TARGET USERS

Organizations where document verification is repetitive, time-consuming and consequential: recruitment, finance, compliance and administrative teams. Initial focus is on teams that already verify documents by hand today and carry the cost of an undetected forgery.

3. INITIAL EVIDENCE

The fraud patterns above are technically observable and were used to shape the system. Two were tested directly against the working prototype:

- An AI-generated invoice, visually flawless, was scored low risk — exposing a blind spot no visual model can close on its own.
- A client tax identifier with an invalid digit count went unnoticed by the model, because counting digits is exactly what vision models fail at.

Both findings changed the product (see §6). User and stakeholder interviews are being conducted to validate current verification workflows and rank the use cases by value.

4. SOLUTION

An AI-assisted screening system that returns not a verdict, but *evidence a human can act on*.

Upload → Normalization → Multimodal analysis → Deterministic checks → Risk score → Explainable report

The key design decision is the division of labour. A vision model is used for what it does well — reading a document and spotting visual manipulation. It is not trusted for what it does badly. Counting digits, verifying a checksum and re-adding a column are handled by deterministic code, whose output is exact and auditable.

Every flagged region is drawn and numbered on the document itself, so the reviewer sees *where* to look, not only *that* something is wrong.

5. PROTOTYPE (BUILT)

A working, deployed web application. What runs today:

- PDF, JPG and PNG upload; the browser renders the document to a single image used both as the preview and as the model payload, which guarantees that flagged regions align with what the reviewer sees.
- Multimodal analysis returning a strict JSON schema, re-validated server-side — a malformed analysis is rejected, never half-displayed.
- Deterministic checks independent of the model: tax identifier length, VAT arithmetic ($HT + VAT = TTC$), date coherence, IBAN mod-97 checksum. A failed check raises the score and blocks an "accept" verdict.
- Output: risk score (0–100), risk level, extracted fields with reading confidence, anomalies with severity and category, numbered regions overlaid on the document, explanation and recommended action.
- Demo mode without an API key, so the interface is always demonstrable.

Deliberately out of scope for 72 hours: OCR-specific tuning, dedicated manipulation-detection models, metadata forensics and multi-document cross-checking. These form the target architecture, not the MVP.

6. VALIDATION

Testing focuses on how documents are verified today, which fraud types are hardest to catch, how long verification takes, and what evidence a reviewer needs to trust an automated result.

Two prototype tests already changed the build: the AI-generated invoice led to the deterministic-check layer, and a contradictory output (low score, reject verdict) led to a consolidation step that keeps score, level and recommendation coherent. A learning log documents who was contacted, what was learned, what was surprising and what changed as a result.

7. BUSINESS LOGIC

Customer. Organizations that verify official documents at volume — lenders, insurers, recruiters, schools, administrations.

Value. Less manual screening, faster identification of suspicious files, and traceable evidence a reviewer can defend.

Model. B2B SaaS per seat or per document, plus an API for integration into existing case-management systems. Pricing to be validated.

Market. Starts with teams having recurring verification needs; expands across finance, recruitment, insurance, education and public administration.

Next 30 days. Interview and pilot with one organization, add metadata forensics and cross-document consistency, extend deterministic checks, produce a timestamped audit report, and test pricing.

KEY ASSUMPTION

Organizations will adopt AI-assisted screening only if it reduces verification effort *and* shows transparent evidence for every flag. A score without justification will not be trusted — and should not be. The product is therefore built around explainability, and states plainly that it is a decision aid, not an expert opinion.